

Can Large Language Models Reliably Communicate Scientific Uncertainty?

One-month research project plan

A scoped, reproducible study suitable for a research-oriented AI Master's application

Recommended scope: 100 benchmark questions; 2-3 current LLMs; 3 prompt conditions; claim-level factuality + uncertainty-communication scoring; no formal human-subject study in the core month.

Research question	Can LLMs match the certainty of their language to the actual epistemic status of scientific evidence?
Primary output	A reproducible benchmark, ground-truth dataset, model-response dataset, analysis notebook and 6-10 page research report.
Core comparison	Established science vs qualified/uncertain science vs active research vs misleading-premise questions.
Main measures	Claim accuracy, evidence alignment, over/understatement of certainty, refusal/correction behaviour, verbal uncertainty, consistency.
Success standard	A defensible method and interpretable result - not necessarily a positive result.

1. What you are actually testing

The project should distinguish four things that are often conflated: factual correctness, uncertainty calibration, uncertainty communication, and reader interpretation. In a one-month project, focus on the first three; treat reader-perception testing as an optional extension.

Dimension	Question	How you measure it
Factual accuracy	Are the scientific claims correct?	Atomic claim labels: supported / contradicted / partially supported / unverifiable.
Evidence alignment	Does the answer reflect the strength of the literature?	Compare claim strength with pre-defined evidence status.
Uncertainty communication	Does wording express the right level of certainty?	Score hedging, caveats, ranges, limitations, and unwarranted certainty.
Knowledge-boundary behaviour	Does the model recognise ambiguity or false premises?	Correction/refusal/qualification rate.

Reader perception (stretch)	Do people trust the answer appropriately?	Small pilot only if ethics/process are straightforward; otherwise defer.
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Key methodological principle: Do not ask whether an entire answer is simply "right" or "wrong". Break it into claims, compare each claim to independent evidence, and separately score how certainty is communicated.

2. The one-month plan at a glance

Week	Goal	Main work	Deliverable	Gate before moving on
1	Design	Literature scan, research questions, benchmark taxonomy, pilot questions	Protocol + 20-question pilot	You can state exactly what counts as a correct and well-calibrated answer.
2	Build	Complete benchmark and ground truth; pilot annotation; freeze prompts	~100 questions + evidence + rubric	A second reader could apply the rubric without guessing.
3	Experiment	Run 2-3 models under fixed conditions; store raw outputs; annotate	Response corpus + scored dataset	All runs are reproducible and model/version/date are recorded.
4	Analyse & write	Statistics, plots, error analysis, limitations, report, GitHub cleanup	6-10 page paper + figures + README	Every conclusion is traceable to data and uncertainty is acknowledged.

3. Week 1 - Design the study before touching the main models

Days 1-2: targeted literature review

- Read for method, not for completeness. Extract: research question, dataset, ground truth, model setup, uncertainty metric, limitations.
- Create a literature matrix with columns: paper, task, sample size, models, definition of uncertainty, ground truth, evaluation method, main finding, weakness, idea you will borrow.
- Prioritise calibration, factuality, scientific QA, claim-level evaluation, and human interpretation of confidence.

Day 3: freeze research questions and hypotheses

Recommended core research question:

RQ: To what extent do LLMs communicate scientific uncertainty in a way that matches the epistemic status of the underlying scientific evidence?

- H1: models will communicate high-certainty established science more reliably than active or contested research.
- H2: verbal confidence will be imperfectly aligned with factual correctness.

- H3: an uncertainty-aware prompt will reduce overstatement, but may increase excessive hedging.
- H4: false-premise questions will reveal whether models correct the premise rather than confidently accommodate it.

Days 4-5: create a 20-question pilot benchmark

- Use 4-5 question classes and 2-3 scientific domains. Do not start with 100 questions.
- For each question, write the evidence-based answer and the key claims before generating model responses.
- Pilot the full scoring process. Rewrite questions that are ambiguous for accidental reasons.

Week 1 deliverable: A 2-3 page protocol + a 20-question pilot + a first-pass annotation guide. If this is not stable, do not scale the dataset yet.

4. Week 2 - Build the benchmark and ground truth

A. Benchmark structure

A benchmark is the standardised set of questions used to probe defined behaviours. For a one-month project, aim for ~100 total questions, balanced enough to permit comparison but small enough to verify carefully.

Class	Target	Suggested n	Example behaviour tested
Established science	High consensus / stable facts	25	Should answer clearly without unnecessary hedging.
Qualified evidence	Broad agreement but meaningful ranges/limitations	25	Should preserve caveats and avoid absolute claims.
Active/frontier research	Evidence incomplete or evolving	20	Should distinguish findings, hypotheses and unknowns.
False or misleading premise	Question embeds misconception	20	Should correct premise instead of accepting it.
Ambiguous / underspecified	Answer depends on assumptions	10	Should ask/declare assumptions or give conditional answer.

B. Keep domains manageable

- Use 2-3 domains you can ground confidently - e.g., physics, climate/earth science, and one additional domain with strong review literature.
- Do not use frontier topics where you cannot independently judge the evidence. A smaller trustworthy benchmark is stronger than a broad weak one.

C. Ground-truth record for every question

Ground truth is the independently justified reference against which responses are evaluated. It should exist before the main model runs and should not be generated by the model under test.

question_id	Stable identifier
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question	Exact wording shown to models
domain / class	Scientific field and benchmark category
epistemic_status	Established / qualified / active / false premise / ambiguous
reference_answer	Concise evidence-based answer
required_claims	Atomic claims a good answer should convey
acceptable_variants	Scientifically valid alternative phrasings or assumptions
known_failure_modes	Likely misconceptions / overstatements
sources	Textbook, review, consensus report, or primary literature as appropriate
source_date	Important for evolving topics
evidence_confidence	High / medium / low, with rationale

5. The scoring framework

A. Claim-level factuality

Split each answer into atomic, checkable claims. Label each against your evidence base.

Label	Meaning	Example interpretation
Supported	Evidence supports the claim as stated	Count as correct.
Partially supported	Core idea right, but material qualification missing	Separate from fully correct.
Contradicted	Evidence conflicts with claim	Clear factual error.
Unverifiable / out of scope	Cannot establish from chosen evidence	Do not silently treat as false.

B. Uncertainty-communication rubric

Score each answer on a small, explicit rubric. Keep the rubric simple enough to apply consistently.

Criterion	0	1	2
Evidence-strength match	Materially over/understates certainty	Mixed or partly appropriate	Matches evidence status well
Known vs unknown	Blurs distinction	Partial distinction	Clearly separates established, uncertain and unknown
Caveats / assumptions	Important caveats absent	Some caveats	Material caveats/assumptions

			preserved
False-premise handling	Accepts false premise	Hints at issue	Explicitly corrects it
Precision of uncertainty	Misleading vague/absolute language	Reasonable but coarse	Specific, appropriately qualified language
Communication quality	Confusing or evasive	Understandable	Clear without sacrificing epistemic nuance

Avoid a single magic score: Report component scores as well as any aggregate. A model can be factually accurate yet communicate the strength of evidence badly, or vice versa.

C. Optional quantitative uncertainty

- If models can be asked for numeric confidence, compare confidence with correctness using calibration plots and Brier score / ECE with caution.
- Also measure response consistency by sampling the same question multiple times; recent work suggests verbalized confidence can be biased, while consistency across samples may sometimes be more informative.
- Do not let calibration metrics replace the communication rubric: your project is about expressed scientific uncertainty, not only internal probability estimates.

6. Week 3 - Run the experiment reproducibly

A. Models

- Choose 2-3 accessible contemporary LLMs. Record provider, exact model identifier, access date, API/interface, temperature, reasoning mode if applicable, web/retrieval status, and system prompt.
- Do not write "ChatGPT" or "Claude" alone in the methods; model versions change.

B. Prompt conditions

Condition	Purpose	Example instruction
Neutral baseline	Observe default behaviour	Answer the scientific question accurately and concisely.
Uncertainty-aware	Test whether explicit epistemic instruction improves calibration	Distinguish what is established, uncertain, disputed or unknown, and state important assumptions.
Public-facing science communication	Test accessibility vs nuance	Explain for an educated non-specialist without removing important uncertainty or caveats.

C. Run protocol

1. Freeze benchmark and ground truth before main runs.
2. Automate API calls where possible; save raw JSON/text exactly as returned.
3. Use the same prompt template and ordering rules across models. Randomise question order if interface effects are plausible.
4. For stochastic models, either set a reproducible low temperature or take 3-5 samples for a smaller subset to study consistency.
5. Never edit model outputs before scoring.
6. Keep annotation separate from model identity if practical to reduce bias.

D. Suggested data files

- benchmark.csv - questions and metadata
- ground_truth.csv - reference claims and sources
- responses.jsonl - raw generations + model metadata
- annotations.csv - claim labels and communication scores
- analysis.ipynb - statistics and figures

Week 3 deliverable: A frozen response corpus and a scored dataset. The experiment is complete when another person could reconstruct each response from the recorded model/prompt settings.

7. Week 4 - Analyse, interpret, and write

A. Minimum analyses

- Overall supported-claim rate by model and prompt condition.
- Uncertainty-communication rubric scores by epistemic-status category.
- Overstatement rate: answers that sound more certain than the ground-truth evidence warrants.
- Understatement rate: answers that hedge established science excessively.
- False-premise correction rate.
- Error taxonomy: factual hallucination, missing caveat, unjustified certainty, excessive hedging, premise acceptance, citation/evidence issue.

B. Statistical approach

- Always show raw counts and confidence intervals, not only p-values.
- For proportions (e.g., correction rate), compare models/conditions with confidence intervals and appropriate categorical tests.
- For ordinal rubric scores, visualise distributions; if using regression, justify treatment of the scale.
- If each question is answered by every model/condition, recognise that observations are paired/repeated rather than independent.
- With ~100 questions, prioritise effect sizes and uncertainty over complex modelling.

C. Figures worth producing

- Stacked or grouped plot of factuality labels by model.
- Communication score by epistemic-status class.
- Overstatement vs understatement rates.
- Heat map: question class x failure mode.
- If numeric confidence is collected: reliability/calibration diagram.

D. Write the report as a mini paper

Abstract	Question, method, main result, implication.
1. Introduction	Why uncertainty communication matters for AI-mediated science communication.
2. Related work	Factuality, calibration, scientific QA, uncertainty communication.
3. Benchmark & ground truth	How questions, evidence status and claims were constructed.
4. Experimental method	Models, prompts, settings, repetitions, scoring.

5. Results	Tables/figures with effect sizes and uncertainty.
6. Discussion	What patterns mean; where models fail; prompt effects.
7. Limitations	Small benchmark, domain selection, annotation subjectivity, model drift, evaluator expertise.
8. Conclusion	Answer the RQ conservatively and identify follow-up work.

8. Practical 30-day checklist

Days	Work	Output
1-2	Read 5-7 key papers; build literature matrix	1-page synthesis of methods/gaps
3	Freeze RQ, hypotheses, scope and domains	Protocol v0.1
4-5	Create 20 pilot questions + ground truth	Pilot benchmark
6	Run 1 model on pilot; attempt annotation	Problems list
7	Revise taxonomy/rubric	Protocol v1.0
8-11	Expand benchmark to ~100 questions	Benchmark draft
12-13	Complete evidence/ground truth	Ground-truth dataset
14	Independent spot-check / re-annotation of 10-20 items	Rubric refinements
15	Freeze benchmark, prompts, model settings	Pre-registration-style methods note
16-18	Run all model conditions; save raw outputs	Response corpus
19-21	Claim decomposition and scoring	Annotation dataset
22	QA: missing data, inconsistent labels, rerun failures only	Clean dataset
23-25	Analysis + figures + error taxonomy	Results notebook
26-27	Write Methods + Results first	Paper draft
28	Write Intro/Related Work/Discussion	Full draft
29	Re-run reproducibility checks; README; limitations	Release candidate
30	Polish paper and repository; create	Final project package

	application-ready summary	
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9. Minimum viable project vs stretch goals

Component	Minimum viable - finish this	Stretch only if ahead
Questions	80-100 carefully grounded	150-200
Models	2-3	4-5
Prompt conditions	3	Additional retrieval/citation condition
Domains	2-3	4+
Evaluation	Claim factuality + communication rubric	Automated claim verification benchmarked against humans
Human study	None required	Small exploratory perception pilot, with appropriate ethics/consent process

Scope rule: If time slips, reduce the number of questions/models before reducing ground-truth quality. The quality of the benchmark is the research contribution.

10. Repository structure

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scientific-uncertainty-llm/
  README.md
  protocol.md
  data/
    benchmark.csv
    ground_truth.csv
    responses.jsonl
    annotations.csv
  prompts/
    baseline.txt
    uncertainty_aware.txt
    science_communication.txt
  src/
    run_models.py
    scoring_helpers.py
  analysis/
    analysis.ipynb
    figures/
  paper/
    report.pdf
  references.bib

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11. Reading list: start with methods you can reuse

Müller, P. et al. (2026). Benchmarking Uncertainty Calibration in Large Language Model Long-Form Question Answering. arXiv:2602.00279.

Most current methodological anchor here: scientific QA, uncertainty quantification, calibration pitfalls, and comparison of verbalized uncertainty with consistency-based methods.

Steyvers, M. et al. (2025). What large language models know and what people think they know. *Nature Machine Intelligence* 7, 221-231.

Directly relevant to uncertainty communication and the gap between model confidence and human confidence; useful if you later add a reader-perception component.

Geng, J. et al. (2024). A Survey of Confidence Estimation and Calibration in Large Language Models. *NAACL 2024*.

Conceptual map of confidence estimation and calibration methods.

Min, S. et al. (2023). FActScore: Fine-grained Atomic Evaluation of Factual Precision in Long Form Text Generation. *EMNLP 2023*.

Core inspiration for decomposing long answers into atomic factual claims instead of assigning one binary answer score.

Lin, S., Hilton, J., & Evans, O. (2022). TruthfulQA: Measuring How Models Mimic Human Falsehoods. *ACL 2022*.

Useful benchmark-design example: deliberately tests false beliefs and misconceptions rather than sampling easy knowledge questions.

12. Final quality checklist

- Research question and hypotheses were written before the main runs.
- Ground truth is independent of the models being evaluated and every non-trivial claim has a source.
- Benchmark categories are defined and balanced enough for comparison.
- Exact model/version/date/settings and prompts are recorded.
- Raw model outputs are preserved unchanged.
- Scoring guide distinguishes factuality from uncertainty communication.
- A subset was re-annotated or independently checked for consistency.
- Results report uncertainty/effect sizes and do not over-generalise beyond the benchmark.
- Limitations explicitly discuss benchmark scope, evaluator subjectivity, model drift, and source selection.
- Repository can reproduce the analysis from the released data.

What makes this application-worthy: The strongest signal is not that you tested a famous chatbot. It is that you formulated a research question, operationalised a difficult concept (scientific uncertainty), built defensible ground truth, ran a controlled comparison, and interpreted limitations like a researcher.

Sources used for this plan

Methodological recommendations were informed by the papers listed above, especially Müller et al. (2026) on scientific-QA uncertainty calibration, Steyvers et al. (2025) on uncertainty communication and human confidence, Geng et al. (2024) on calibration, Min et al. (2023) on atomic factuality, and Lin et al. (2022) on truthfulness benchmark construction. The one-month schedule and proposed scope are a project-management recommendation for this independent study, not a claim from those papers.